

VibeHaus — Architecture & Technical Overview (Simple Explanation)

Author: Sakshi · Capstone · *A plain-language overview, not an engineering spec.*

The stack, in one sentence

VibeHaus is a **mobile-first responsive web app** (not a native iOS/Android app — see backlog, Epic J2, Future), built with **React** on the frontend, backed by **Supabase** for authentication and data storage, built and deployed via **Lovable**.

How a request flows through the app

User's phone/browser

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React frontend (the screens: Discover, My Haus, Chat, Events, Calendar...)

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Supabase (auth + database) — stores profiles, Vibemate relationships, messages, Coffee/Activity Vibes, Events, and their statuses

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Two AI touchpoints, called only where needed:

1. Supervised ML model → ranks compatibility (Vibe Match scoring)
2. LLM → writes the "why you might vibe" explanation, Icebreakers, and Mr. Vibe's scripted replies

The three-layer AI design (why it's built this way)

1. **Data layer** — profile fields the user actually provided: interests, social energy, intent, availability, connection preference. No protected attributes, no ethnicity, no payment/ID data in the MVP.
2. **Supervised ML layer** — a rule/attribute-weighted model decides **who** ranks as compatible with whom. This is deterministic and explainable by design, not an opaque black box.

3. **LLM layer** — used **only** to turn the ML's ranking into plain-language text: the "why you might vibe" explanation, Icebreakers, and Mr. Vibe's replies. The LLM never decides the match itself, it only explains a decision the rule-based layer already made.
4. **Scripted scaffolding** — Mr. Vibe's actual conversation flow (greetings, chip menus, fallback messages) is fixed, pre-written text, not live-generated. This was a deliberate choice, made after testing revealed a live-generative version was unreliable (see the AI Responsibility Statement) — reliability and cost control mattered more than open-ended flexibility for this feature.

Why this matters (the honest trade-off)

A fully generative, free-text AI throughout would be flashier, but riskier and slower to get right under a five-day sprint. Using **ML for the decision, LLM for the explanation, and scripting for reliability** gave a system that's transparent (every match has a stated reason), predictable (nothing the user sees is a live gamble), and fast to build correctly — at the cost of less open-ended flexibility than a fully free-text assistant would offer. That trade-off is documented, not hidden, and it's the same reasoning behind Mr. Vibe's chip-only design.

What's static vs. live right now (MVP honesty)

- **Static (not yet live):** venue maps show a fixed preview image, not an interactive map (Epic N1, Next phase). Identity verification is not implemented; no "Verified" badge is shown, to avoid implying a check that isn't performed.
- **Live:** authentication, profile data, Vibemate relationships, chat messages, Events, and Calendar are all real, persisted data via Supabase — not mocked, except for the seeded demo profiles used to populate Discover for testing.